



DATA-500: INTRODUCTION TO AN OPEN SOURCE FUTURE: DECOLONIZING TECHNOLOGY & DATA SOVEREIGNTY



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ABSTRACT

This FacultyHack project redesigned Introduction to an Open Source Future: Decolonizing Technology and Data Sovereignty at Howard University to integrate SGX3 cyberinfrastructure utilizing Jetstream2 virtual machines, multi-user JupyterHub, and Podman containers within a course long, open dataset-driven project theme. Students work with real environmental justice and anti-surveillance datasets across the semester progressing from Linux operating system fundamentals and data exploration in JupyterLab and Metabase to containerized workflows with Podman and local open-source AI tools via Jetstream2 supplies the persistent computing environment and JupyterHub instance that allow every student, regardless of personal hardware, to run the same reproducible stack, while Podman reinforces the course's commitment to secure, rootless, open-source tooling.

The redesign addresses a concrete gap: many students enter with limited Linux operating system and open source tool or command-line experience yet have strong interest in questions of data sovereignty and digital power. By anchoring every technical skill to analysis of the same evolving dataset, the course converts abstract linux tool based learning into purposeful environmental justice data analytics practice. Students leave proficient in their ability to install and manage Linux environments, containerize analysis pipelines, query and visualize data, and critically evaluate the infrastructures that shape knowledge production. The approach builds genuine confidence in open-source tools while preparing learners for subsequent courses in data science, AI automation, and environmental sensing. Two practical lessons emerged once target level is achieved. First, a single, well-chosen dataset sustained across modules creates stronger continuity and motivation than a sequence of disconnected exercises; students repeatedly return to the same data with new tools and deeper questions. Second, introducing Jetstream2 and JupyterHub early as the shared classroom workspace in addition to bare-metal optional builds, rather than as an advanced add-on, normalizes national cyberinfrastructure as an everyday open-source resource and reduces the intimidation that often accompanies first encounters with HPC platforms. When combined with proficiency in local open source tools, national cyberinfrastructure system broadens the capability of students regardless of budget.

INTRODUCTION

Many students enter with limited Linux and command-line experience yet have strong interest in data sovereignty and the power structures of big tech and data sovereignty. The existing curriculum emphasizes Python, R, Tableau, and GIS without systematically teaching open-source operating systems, containers, or national cyberinfrastructure. This gap leaves learners dependent on proprietary platforms and less equipped to build fully reproducible, autonomous analytical environments. Integrating Jetstream2 virtual machines, multi-user JupyterHub, and Podman containers through a continuous environmental-justice and anti-surveillance dataset project was therefore motivated by the need to give every student practical confidence in open-source tools and a clear technical pathway into advanced data science and ethical technology practice.

Course Goals

- Master core concepts of open-source operating systems, local tooling, and ethical computing practices that challenge digital colonialism and advance data sovereignty.
- Develop computational problem-solving skills by applying open source and containerized workflows to progressive analysis of environmental justice and anti-surveillance datasets.
- Emphasize professional best practices in documentation, reproducibility, and collaborative work through shared JupyterHub environments and version-controlled analysis pipelines.
- Gain practical proficiency with SGX3 cyberinfrastructure by using Jetstream2 virtual machines, multi-user JupyterHub, and Podman containers as the persistent, accessible computing foundation throughout the course.

Grade Computation

Problem Sets	50%
Memo/Briefings	20%
Lightening Talks/Slides	20%
Final Poster	10%
Participation/Bonus	5%

Mentor Feedback Incorporated

- Defined dataset to be used throughout the course
- Define level 500 and audience; set up resources for 20. NAIRR application.
- Curate and focus lecture around theme and SGX3 CyberInfrastructure resources

IMPACT ON STUDENTS

Who benefited, and what changed for them.

Who are our users — students in this course, a program, a broader community? What changed for them: skills gained, confidence with computing, access to resources they couldn't reach before? Include a representative observation if you have one.

COURSE REDESIGN: BEFORE & AFTER

INITIAL SYLLABUS

- The course covered a wide range of open source tools utilizing a range of datasets. Readings, and self directed projects
- Students we previously only given the option to install operating system on an old laptop or tablet.
- What was missing** — A cross-cutting theme or dataset underlying the progression of the lessons and assignments
 - Limiting options of installation of software and operating system environment presented logistical difficulties as well as

MODIFIED SYLLABUS

- Added Deflock and related datasets for overlay while modifying the grading structure for simplicity and accessibility
- HPC now integrated throughout the course with parallel options for software interfacing and workflow development.
- The assessment changed to produce a more accessible course providing students with optionality that includes ACCESS hosted solutions

TARGET COURSE DESCRIPTION

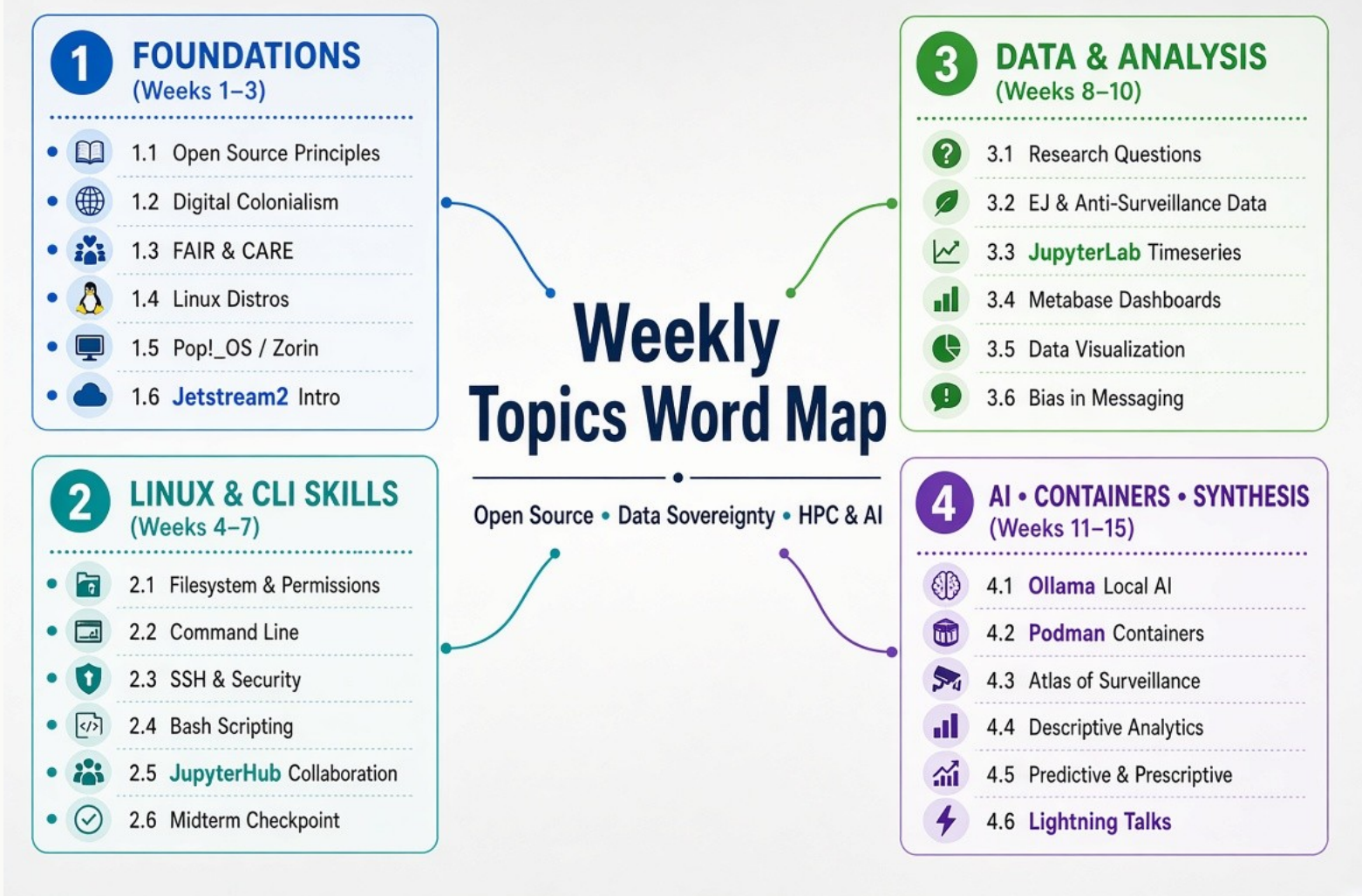
Catalog-style facts: name, number, department, enrollment, prerequisites, repo.

Course Name	D500
Course Number	DATA-500
Department	Applied Data Science and Analytics
Enrollment	15-25
Prerequisites	DATA-500, Anticipate Fall & Spring
Project Repo	github.com/your-org/your-repo

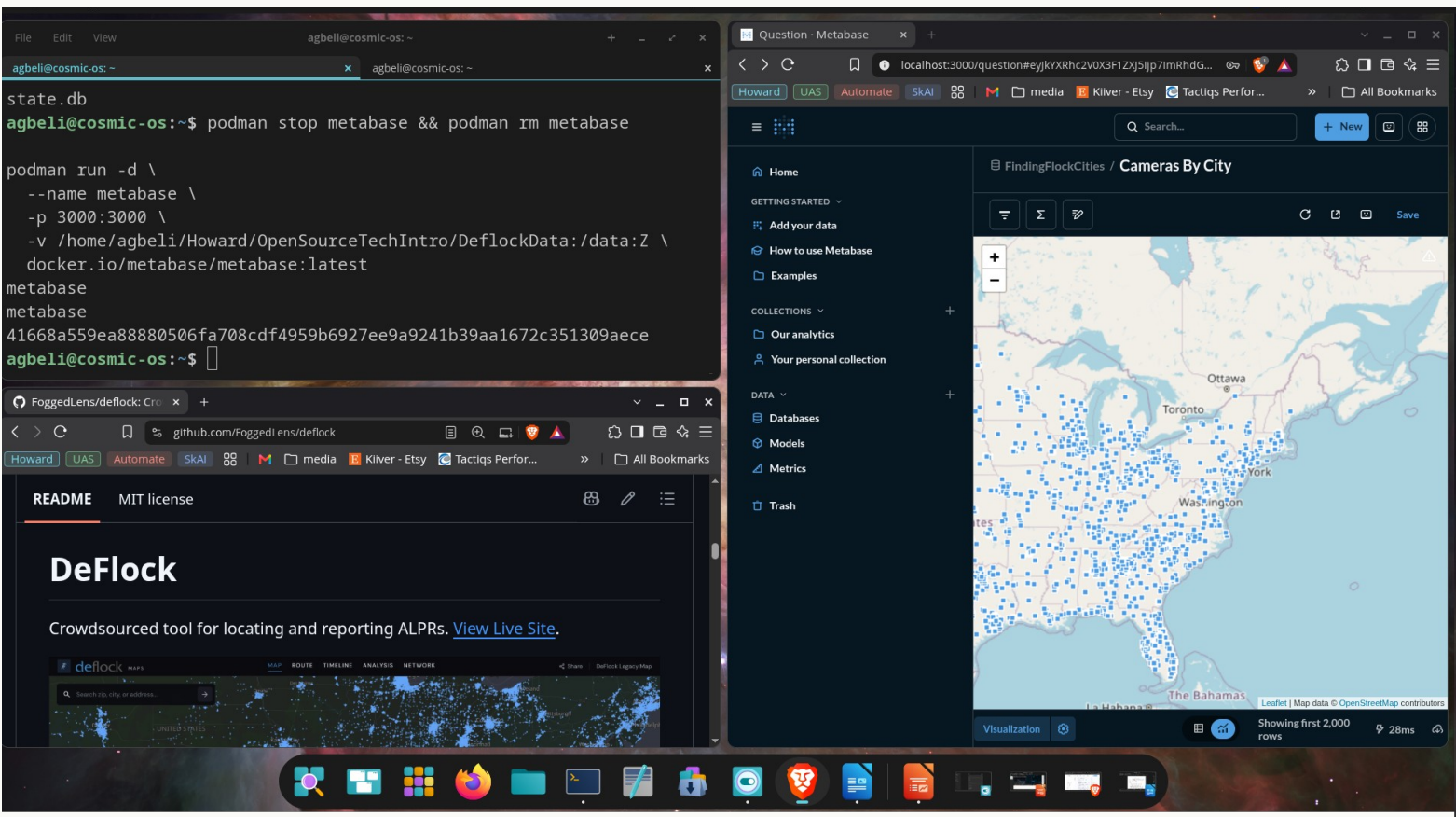
Course Description

A hands-on course at Howard University that equips students with practical open-source skills—Linux (Pop!_OS/Zorin), JupyterLab, Metabase, local AI (Ollama), and automation tools—while critically examining digital colonialism and data sovereignty. Students learn to reclaim computing autonomy, analyze environmental justice datasets, and propose ethical open-source solutions, supported by Science Gateways and Jetstream2 resources.

Key Content by Week



SCIENCE GATEWAY RESOURCES USED



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PROJECT GITHUB REPO

github.com/Teanghana/HUDATA500

Sponsors: SGX3 · National Science Foundation · ORNL · Omnibond · TACC · HackHPC



FacultyHack @Gateways26
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